SECTION TABLE OF CONTENTS

DIVISION 31 - EARTHWORK

SECTION 31 00 00

EARTHWORK

-- End of Section Table of Contents --

SECTION 31 00 00

EARTHWORK

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C136/C136M	(2019) Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates
ASTM D698	(2012; E 2014; E 2015) Laboratory Compaction Characteristics of Soil Using Standard Effort (12,400 ft-lbf/cu. ft. (600 kN-m/cu. m.))
ASTM D1140	(2017) Standard Test Methods for Determining the Amount of Material Finer than 75-μm (No. 200) Sieve in Soils by Washing
ASTM D1556/D1556M	(2015; E 2016) Standard Test Method for Density and Unit Weight of Soil in Place by Sand-Cone Method
ASTM D2216	(2019) Standard Test Methods for Laboratory Determination of Water (Moisture) Content of Soil and Rock by Mass
ASTM D2321	(2020) Standard Practice for Underground Installation of Thermoplastic Pipe for Sewers and Other Gravity-Flow Applications
ASTM D2487	(2017; E 2020) Standard Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System)
ASTM D2974	(2020; E 2020) Moisture, Ash, and Organic Matter of Peat and Other Organic Soils
ASTM D6913/D6913M	(2017) Standard Test Methods for Particle-Size Distribution (Gradation) of Soils Using Sieve Analysis
ASTM D6938	(2017a) Standard Test Method for In-Place Density and Water Content of Soil and Soil-Aggregate by Nuclear Methods (Shallow Depth)

FLORIDA DEPARTMENT OF TRANSPORTATION (FDOT)

FDOT Specifications	(2025-26) Standard Specifications for Road and Bridge Construction

U.S. ARMY CORPS OF ENGINEERS (USACE)

GDR	SECTION 00 31 32, Geotechnical Data Report for INDIAN RIVER LAGOON SOUTH (IRL-S), C-44 RESERVOIR AND STORMWATER TREATMENT AREA (C-44 RSTA), SEEPAGE MANAGEMENT FEATURE - RELIEF WELLS CONTRACT A
EM 385-1-1	(2024) Safety and Occupational Health (SOH) Requirements

1.2 DEFINITIONS

1.2.1 Topsoil

Surface layer of soil that is currently supporting vegetation growth in the area of work and is capable of supporting the reestablishment and replacement of ground cover required by Section 32 92 23 SODDING.

1.2.2 Pipe Bedding

Fill placed to directly support pipes, conduits, cables, and appurtenant structures. Bedding may also be used to provide a cushion between utilities and bedrock, obstacles, obstructions and other unyielding materials.

1.2.3 Flowable Fill

Fill placed in a plastic or liquid form that flows to near its final placement location with limited assistance and subsequently cures or solidifies to provide a stable or impermeable barrier.

1.2.4 Satisfactory Materials

Satisfactory materials for fill, backfill, and/or any in-situ soils to remain in place comprise any materials classified by ASTM D2487 as SP, SM, SC, SP-SM, SP-SC, have a maximum organic content of 3 percent when tested in accordance with ASTM D2974, and are not considered unsatisfactory. Fill and backfill is generated from borrow and must have a maximum particle size no greater than 3 inches.

1.2.5 Unsatisfactory Materials

Materials which do not comply with the requirements for satisfactory materials are unsatisfactory. Unsatisfactory materials also include man-made fills; trash; refuse; roots and other organic matter.

1.2.6 Unstable Material

Unstable materials are too weak to adequately support the utility pipe, conduit, equipment, or appurtenant structure. Satisfactory material may become unstable due to ineffective drainage, dewatering, excessive loading, or vibration.

1.2.7 Degree of Compaction (Proctor)

Degree of compaction required is expressed as a percentage of the maximum density obtained by the test procedure presented in ASTM D698 abbreviated as a percent of laboratory maximum density.

1.2.8 Borrow

The soils generated after stripping and originating from the excavations necessary for constructing the manhole and outfall shown on the drawings.

1.3 SUBSURFACE DATA

Subsurface information and data is included in Section 00 31 32 GEOTECHNICAL DATA REPORT (GDR), which contains important information for bidding the work. Subsurface soil boring logs are indicated in the GDR. These data represent relevant subsurface information in the vicinity of the work; however, variations exist between boring locations. The groundwater level in the construction limits is variable and is heavily influenced by the reservoir stage. Artesian pressures exist within the construction limits. Refer to Section 00 31 32 GEOTECHNICAL DATA REPORT for groundwater elevation readings for relevant piezometers and monitoring wells and important information on the subsurface and groundwater conditions. During the work, the reservoir will be operated up to elevation 36 feet NAVD88. Refer to Section 00 33 50 WEATHER AND WATER STAGE DATA for important information on reservoir operations.

1.4 CRITERIA FOR BIDDING

Base bids on the following criteria:

a. Surface elevations are as indicated and the work is performed in accordance with this section.

b. Pipes or other artificial obstructions, except those indicated, will not be encountered.

c. Anticipate that each relief well will flow freely at the top of riser (TOR) elevation defined in the drawings and the reservoir will be operated as described in Section 00 33 50 WEATHER AND WATER STAGE DATA.

d. Material character is as documented in the GDR.

e. Assume 20 percent of material encountered in the required excavations or at the lines and grades indicated will have to be wasted or improved due to organic content or fines content and excess material is wasted.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Excavation and Trenching Plan; G, DO

Borrow Plan; G, DO

Dewatering Work Plan; G, DO

SD-03 Product Data

Flowable Fill Mix Design; G, RO

Geotextiles

SD-04 Samples

Geotextiles

SD-06 Test Reports

Material Test Report; G, RO

Pipe Inspection Report; G, RO

1.6 QUALITY CONTROL

1.6.1 Geotechnical Engineer

Provide a Professional Geotechnical Engineer to provide inspection of excavations and soil/groundwater conditions throughout construction. The Geotechnical Engineer is responsible for the Dewatering Work Plan and any updates necessary as construction progresses.

1.6.2 Qualified Technician

Provide a Qualified Technician to inspect, monitor, sample, and performing field testing. The technician qualifications need to be one of the following: a current National Institute for Certification in Engineering Technologies (NICET) Level II minimum certification in Construction Materials Testing Soils; a Geologist-in-Training with minimum one-year experience; an Engineer-in-Training with minimum one-year experience; a Registered Geologist; or a Professional Engineer.

1.6.3 Lab Validation

Perform testing by a Corps validated commercial testing laboratory or Contractor established testing laboratory approved by the Contracting Officer. Submit testing laboratory validation for the testing to be performed. Do not permit work requiring testing until testing facilities have been inspected, Corps validated and approved by the Contracting Officer.

PART 2 PRODUCTS

2.1 SOIL MATERIALS

2.1.1 Fill

In accordance with the definition of Satisfactory Materials.

2.1.2 Topsoil

Provide topsoil in accordance with the definition of topsoil.

2.2 PIPE BEDDING

Provide bedding for buried piping in accordance with ASTM D2321 and below. Install bedding for plastic piping from 4 inches below to springline of pipe. Provide geotextile fabric below bedding layer when bedding gradation does not meet "Migration" requirements of ASTM D2321, Appendix X1. Bedding material may include the following:

2.2.1 Gravel and Crushed Stone

Clean, coarsely graded crushed limestone meeting the gradation and quality requirements of No. 89 Stone of FDOT Specifications Section 901 or having a classification of GW in accordance with ASTM D2487 for bedding as indicated. Do not exceed maximum particle size of 1-1/2 inches. The No. 89 gradation does not meet "Migration" requirements of ASTM D2321.

2.3 FLOWABLE FILL

Flowable Fill may be used for outfall construction, such as around or under the manhole base slab, outfall pipe, or endwall. Flowable fill is required to fill any voids beneath precast concrete elements (i.e. the manhole base slab, endwall, etc.). Provide flowable fill from an approved FDOT source that meets the requirements of FDOT Specifications Section 121 for Non-Excavatable flowable fill, and submit the flowable fill mix design and depict where it will be used.

2.4 GRAVEL

Provide gravel for pipe outfall from an FDOT approved source and in accordance with No. 1 stone gradation requirements of FDOT Specifications Section 901.

2.5 BORROW

Provide borrow materials meeting the requirements of paragraph SOIL MATERIALS above.

2.6 GEOTEXTILE

Provide non-woven geotextile fabric conforming to the requirements of FDOT Specifications Sections 985, Type D-4 and at least a mass per unit area of 12 oz per square yard. Submit a sample and material product data for all Geotextiles utilized.

PART 3 EXECUTION

3.1 PROTECTION

Perform all work specified in accordance with applicable requirements of the Corps of Engineers publication EM 385-1-1 Safety and Health Requirements Manual.

Use equipment of type and size appropriate for the site conditions (soil character and moisture content). Maintenance of exposed subgrades and fills is the responsibility of the Contractor. The Contractor is required to prevent damage by ineffective drainage, dewatering, and heavy loads and equipment by implementing precautionary measures. Repair or replace any defects or damage.

3.1.1 Underground Utilities

Location of the existing utilities indicated is approximate. When working in the vicinity of utilities, including buried piping, piezometers, and monitoring wells, physically verify the location and elevation of the existing features indicated prior to starting construction. Protect utilities from damage during construction, and repair any damage caused by construction activities.

3.1.2 Drainage and Dewatering

Provide for the collection and disposal of surface and subsurface water encountered during construction.

3.1.2.1 Drainage

Provide for the collection and disposal of surface and subsurface water encountered during construction. Maintain and provide positive surface water runoff away from the construction activity or provide temporary ditches, swales, berms, silt fencing or other drainage features and equipment to keep soils from becoming unstable, prevent erosion, or interfering with the work. Remove unstable material from working platforms for equipment operation and soil support for subsequent construction features and provide new material as specified herein. It is the responsibility of the Contractor to assess the site conditions to employ necessary measures to permit construction to proceed.

3.1.2.2 Dewatering

Control groundwater flowing toward or into excavations to prevent sloughing of excavation slopes and walls, boils, uplift and heave in the excavation and to eliminate interference with orderly progress of construction. French drains, sumps, ditches or trenches are not allowed within 5 feet of the foundation of any manhole, except with specific written approval from the Government, and after specific contractual provisions for restoration of the foundation area have been made. Active pumping of the relief wells is not restricted. Perform control measures by the time the excavation reaches the water level to maintain the integrity of the in-situ material. Initiate dewatering efforts prior to excavation. While the excavation is open, maintain the water level continuously, at least 2 feet below the working level. Submit a Dewatering Work Plan outlining procedures for accomplishing dewatering work and include any effects this may have on the WRPC water level. Lowering the WRPC water level below El 18 will not be allowed. Operate dewatering system continuously until construction work below existing water levels is complete. Do not allow water to seep up along the annulus of the relief well or grout seal and into the excavation. The excavation will not be considered dewatered if the subgrade is pumping or water is seeping into the exvavation. The contractor is responsible for identifying and obtaining any necessary permits for dewatering.

3.1.3 Shoring and Sheeting

Submit an Excavation and Trenching Plan to stabilize features, prevent undermining or unintended horizontal and vertical movement of adjacent structures, and prevent slippage or movement in banks or slopes adjacent to the excavation. Submit drawings and calculations, certified by a registered professional engineer, describing the methods for shoring and sheeting of excavations. Drawings to include material sizes and types, arrangement of members, and the sequence and method of installation and removal. Calculations are to include data and references used. The use of sheeting that, when installed or removed, will disturb the existing drain sand in the perimeter canal will not be permitted.

3.1.4 Protection of Graded Surfaces

Protect newly backfilled, graded, and topsoiled areas from traffic, erosion, and settlements that may occur. Repair or reestablish damaged grades, elevations, or slopes.

3.2 BORROW

Obtain borrow material in accordance sub-paragraph "Borrow" of paragraph DEFINITIONS. Segregate satisfactory materials from unsatisfactory materials. Stockpile all satisfactory materials. Thoroughly mix borrow and moisture condition to meet the requirements and conditions of the fill to be used. Dispose of all unsatisfactory materials at the locations indicated on the plans. Submit a Borrow Plan that includes materials to be excavated, stockpile locations, provisions for thorough mixing, moisture conditioning, and segregation and disposal of unsatisfactory materials.

3.2.1 Stripping and Stockpiling Operations

Strip in accordance with paragraph SURFACE PREPARATION below. Strip at least 5 feet beyond the limits of the borrow excavation and any stockpiles of fill materials.

Stockpile materials within the borrow area work limits such that the stockpiles do not interfere with the work. Stockpile borrow material awaiting transport in approved segregated piles.

3.3 SURFACE PREPARATION

3.3.1 Stripping

Strip existing surface materials to a depth of 6 inches below the existing ground surface. Segregate stripped materials not suitable for reuse as topsoil. Stockpile stripping materials suitable for use as topsoil separately from other excavated material. Dispose of stripped material not suitable for use as topsoil in the area designated on the drawings. Protect topsoil and keep in segregated piles until needed.

3.3.2 Stockpiling Operations

Place and grade stockpiles of materials as specified. Keep stockpiles in a neat and well drained condition, giving due consideration to drainage at all times. Clear, grub, and seal by roller or rubber-tired equipment, the ground surface at stockpile locations; separately stockpile materials by type. Protect stockpiles of satisfactory materials from contamination which may destroy the quality and fitness of the stockpiled material. Do not create stockpiles that could obstruct the flow of any stream, endanger a partly finished structure, impair the efficiency or appearance of any structure, or be detrimental to the completed work in any way. If the

Contractor fails to protect the stockpiles, and any material becomes unsatisfactory, remove and replace such material with satisfactory material from approved sources.

3.4 EXCAVATION

Excavate to contours, elevation, and dimensions indicated. Excavate soil disturbed or weakened by Contractor's operations, and soils softened or made unstable for subsequent construction due to exposure to weather. Use material removed from excavations meeting the specified requirements in the construction of fills or backfill, and for similar purposes to minimize surplus material and to minimize additional material to brought on site. Do not excavate below indicated depths except to remove unstable material as determined by the Government and confirmed by the Contracting Officer. Remove and replace excavations below the grades shown with appropriate materials as directed by the Contracting Officer.

If at any time during excavation, including excavation from borrow areas, the Contractor encounters material that may be classified as rock or as hard/unyielding material, uncover such material, and notify the Contracting Officer. Do not proceed with the excavation of this material until the Contracting Officer has classified the materials as common excavation or rock excavation. Failure on the part of the Contractor to uncover such material, notify the Contracting Officer, and allow sufficient time for classification and delineation of the undisturbed surface of such material will cause the forfeiture of the Contractor's right of claim to any classification or volume of material to be paid for other than that allowed by the Contracting Officer for the areas of work in which such deposits occur.

3.4.1 Trench Excavation Requirements

Excavate the trench as recommended by the manufacturer of the pipe to be installed. Do not exceed the trench width of 24 inches below the top pipe plus pipe outside diameter (O.D.) for pipes of less than 24 inches inside diameter. Neatly cut and dispose of existing Turf Reinforcing Mat (TRM) and any associated pins or anchors, as indicated on the drawings.

3.4.1.1 Bottom Preparation

Grade the bottoms of trenches accurately along the full length of pipe.

3.4.1.2 Removal of Unstable Material

Where unstable material is encountered in the bottom of the trench, remove such material to the depth directed and replace it to the proper grade with suitable material as provided in paragraph FILLING AND COMPACTION. When removal of unstable material is required due to the Contractor's fault or neglect in performing the work, the Contractor is responsible for excavating the resulting material and replacing it without additional cost to the Government.

3.4.1.3 Excavation for Appurtenances

Provide excavation for manholes, catch-basins, inlets, or similar structures sufficient to leave at least 12 inches clear between the outer structure surfaces and the face of the excavation or support members. Do not begin excavation until dewatered and all materials are available and ready to be placed.

3.4.2 Underground Utilities

Excavation made with power-driven equipment is not permitted within 2 feet of known utility or subsurface construction. For work immediately adjacent to or for excavations exposing a utility or other buried obstruction, excavate by hand. Start hand excavation on each side of the indicated obstruction and continue until the obstruction is uncovered or until clearance for the new grade is assured. Support uncovered lines or other existing work affected by the contract excavation until approval for backfill is granted by the Contracting Officer. Report damage to existing subsurface construction immediately to the Contracting Officer.

3.5 SUBGRADE PREPARATION

3.5.1 General Requirements

Shape subgrade to line, grade, and cross section as needed to access the excavation. Remove unsatisfactory and unstable material in surfaces to receive fill or in excavated areas, as determined by the Contracting Officer and replaced with satisfactory materials. Do not place material on surfaces that are muddy, or contain unacceptable materials or unstable material.

3.5.2 Subgrade for Appurtenances

Do not excavate to final grade until the temporary casing, manhole and initial backfill can be completed in the same day. Do not excavate below depth shown for manholes. If over excavation occurs, notify the Contracting Officer and remove, replace, and compact to a minimum of 95 percent of ASTM D698 and firm and unyielding surface, as directed.

3.5.3 Subgrade Filter Fabric

Place filter fabric as indicated directly on prepared subgrade free of vegetation, stumps, rocks larger than 2 inch diameter and other debris which may puncture or otherwise damage the fabric. Repair damaged fabric by placing an additional layer of fabric to cover the damaged area a minimum of 3 feet overlap in all directions. Overlap fabric at joints a minimum of 3 feet. Obtain approval of filter fabric installation before placing fill or backfill. Place gravel on fabric in the direction of overlaps and compact as specified herein. Follow manufacturer's recommended installation procedures.

3.6 FILLING AND COMPACTION

Prepare ground surface on which backfill is to be placed and provide compaction requirements for backfill materials in conformance with the applicable portions of paragraphs for SUBGRADE PREPARATION. Finish compaction by hand operated compaction equipment. Moisten material as necessary to provide the moisture content that will readily facilitate obtaining the specified compaction with the equipment used. Fill and backfill to contours, elevations, and dimensions indicated. Compact and test each lift before placing overlaying lift.

3.6.1 Trench Backfill

Backfill trenches to the grade indicated.

3.6.1.1 Replacement of Unstable Material

Replace unstable material removed from the bottom of the trench or excavation with satisfactory material placed in layers not exceeding 6 inches loose thickness.

3.6.1.2 Bedding and Initial Backfill

Provide bedding of the type and thickness indicated place accurately to within 0.05 ft. of grade on which the pipe rests. Place initial backfill material and compact it with approved tampers to a height of at least one foot above the utility pipe or conduit. Bring up the backfill evenly on both sides of the pipe for the full length of the pipe. Take care to ensure thorough compaction of the fill under the haunches of the pipe. Except where shown or when specified otherwise, provide bedding for buried piping in accordance with PART 2 paragraph PIPE BEDDING. Compact backfill to top of pipe to 95 percent of ASTM D698 maximum density.

3.6.1.3 Final Backfill

Do not begin until pipes have been inspected, tested and approved, and the excavation cleaned of trash and debris. Bring backfill to indicated finish grade. Compact remaining area in layers not more than 4 inches in compacted thickness with power-driven hand tampers suitable for the material being compacted. Where the existing TRM was cut, key in the TRM as shown on the drawings and compact backfill.

3.6.1.4 Displacement of Features

After other required tests have been performed and the trench backfill compacted to 2 feet above the top of the pipe, inspect the pipe to determine whether unexpected or damaging displacement has occurred. Inspect pipes smaller than 48 inches using remote methods using closed circuit television, sonar, or hybrid that can provide a 360-degree inspection of the pipe. Prepare and submit a pipe inspection report consisting of digital video or photos. If, in the judgment of the Contracting Officer, the interior of the pipe shows poor alignment or any other defects that would cause improper functioning of the system, replace or repair the defects as directed at no additional cost to the Government.

3.6.2 Fill Placement

Place fill and backfill beneath and adjacent to structures in successive horizontal layers of loose material not more than 8 inches in depth, or in loose layers not more than 4 inches in depth when using hand-operated compaction equipment. Do not place over wet materials. Compact to at least 95 percent of laboratory maximum density, except as otherwise specified. Perform compaction in such a manner as to prevent wedging action or eccentric loading upon or other damage to the structure. Moisture condition fill and backfill material to a moisture content that will readily facilitate obtaining the specified compaction. Roughen level surfaces for fill or backfill. Cut sloped surfaces into rough steps or benches, or roughen sloped surface to provide a satisfactory bond for compacting materials.

3.6.3 Backfill for Appurtenances

After the manhole or endwall has been placed, place backfill in such a manner that the structure is not be damaged or displaced by the shock of falling earth. Deposit the backfill material, compact it as specified for fill placement, and bring up the backfill evenly on all sides of the structure to prevent eccentric loading and excessive stress.

3.6.4 Flowable Fill

Place fill in a manner to completely fill voids in the location indicated. Do not place when atmospheric temperatures are expected to be below 33 degrees F at any time during the 3 day period following placement.

3.6.5 Turfed or Seeded Areas Miscellaneous Areas

Deposit backfill in layers of a maximum of 12 inches loose thickness, and compact it to 90 percent maximum density. Do not permit compaction by water flooding or jetting. Apply this requirement to all other areas not specifically designated above.

3.7 GRAVEL APRON

Construct gravel apron on filter fabric in the areas indicated. Trim and dress indicated areas to conform to cross sections, lines and grades shown with a minimum thickness as shown on the plans.

3.7.1 Gravel Placement

Place gravel for apron to produce a well graded mass with the minimum practicable percentage of voids in conformance with lines and grades indicated. Place gravel on geotextile without damage and only when the subgrade is dry and stable. Distribute larger rock fragments, with dimensions extending the full depth of the rip-rap throughout the entire mass and eliminate "pockets" of small rock fragments. Rearrange individual pieces by mechanical equipment or by hand as necessary to obtain the distribution of fragment sizes specified above. Hand-place surface rock with open joints to create a final surface that is stable and can be safely walked upon.

3.8 FINISHING/FINISH OPERATIONS

3.8.1 Grading

Finish grades as indicated within 0.1 foot and grade areas to drain surface water runoff as indicated on the plans and without abrupt changes in grade or water ponding. Maintain areas free of trash and debris. For existing grades that will remain but which were disturbed by Contractor's operations, grade as directed.

3.8.2 Topsoil

On areas to receive topsoil, prepare the compacted subgrade soil to a 2 inches depth for bonding of topsoil with subsoil. Spread topsoil evenly to a thickness of 6 inches and grade to the elevations and slopes shown. Do not spread topsoil when excessively wet or dry. Keep topsoil separate from other excavated materials, brush, litter, objectionable weeds, roots, stones larger than 2 inches in diameter, and other materials that would interfere with planting and maintenance operations.

3.9 DISPOSITION OF SURPLUS MATERIAL

Remove surplus soil material. Dispose of surplus soil material in the area indicated on the drawings. Keep satisfactory , unsatisfactory, and topsoil materials in separate stockpiles. Remove from Government property any other trash, debris, or other material generated from the work. This includes any TRM, geotextile, flowable fill, etc.

3.10 TESTING

Perform testing as indicated in Table 1. Submit Material Test Reports within 24 hours of tests being completed.

Material Type	Location of Material	Test Method	Test Frequency
Fill	Manhole Excavaton Backfill	In-place Density - ASTM D6938. When ASTM D6938 is used, check the calibration curves and adjust using only the sand cone method as described in ASTM D1556/D1556M.	One test for each manhole subgrade and each lift of fill. Check in-place densities by ASTM D1556/D1556M as follows: One check test per manhole.
Fill	Blended Manhole Stockpile	Moisture Density Relationship - ASTM D698. Classification - ASTM D2487. Moisture Content - ASTM D2216. Organic Content - ASTM D2974. Gradation - ASTM D6913/D6913M. Fines Content - ASTM D1140.	One representative test per each blended manhole stockpile.
Pipe Bedding	Stockpile	Gradation - ASTM C136/C136M	One representative test per truckload. Sample to be taken at time of delivery.

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